

Final Report: Mach-Zehnder Interferometer

UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis

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I. INTRODUCTION

A Mach-Zehnder Interferometer (MZI) is a fundamental component widely used in photonic integrated circuits (PICs). It operates by splitting incoming light into two separate optical paths, which introduces a relative phase difference between them, and then recombining the signals to produce interference. In a typical PIC implementation, the MZI consists of two optical splitters (often Y-branches, directional couplers, or multimode interferometers) connected by two waveguide arms. By engineering the optical path length difference between these arms (ΔL), a wavelength-dependent periodic interference pattern is generated. The spacing between adjacent transmission peaks, known as the free spectral range (FSR), is directly related to this path length difference. Additionally, the measured FSR provides a convenient way to extract the group index of the waveguide.

II. MODELING AND SIMULATION

A. Waveguide

The waveguide used in this study is a silicon core strip waveguide with a width of 500 nm and a height of 220 nm, surrounded by SiO_2 cladding (Figure 1).

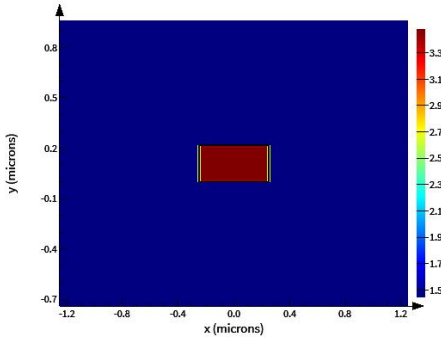


Fig. 1. The cross-section of the waveguide.

Lumerical MODE solver is used to obtain the E-field intensity profile for the fundamental TE and TM mode, as shown in Figure 2 and Figure 3, respectively.

A wavelength sweep from 1500 nm to 1600 nm is performed to determine the relationships between the effective index and wavelength, as well as between the group index and wavelength. For TE mode, the effective index decreases as the wavelength increases (Figure 4), while the group

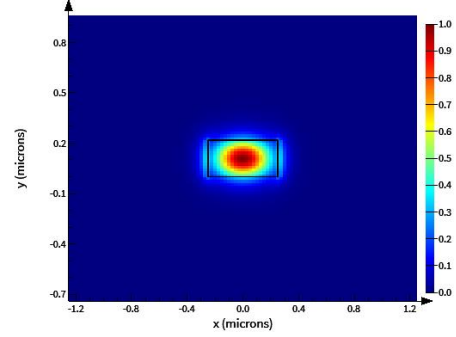


Fig. 2. The simulated waveguide mode profile for TE polarization.

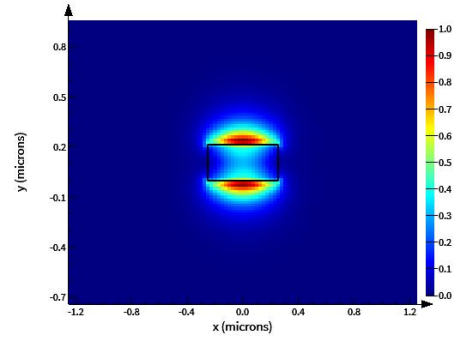


Fig. 3. The simulated waveguide mode profile for TM polarization.

index increases as the wavelength increases (Figure 5). For TM mode, the effective index decreases as the wavelength increases (Figure 6), and the group index also decreases as the wavelength increases (Figure 7).

A compact waveguide model can be obtained by fitting the simulated data to a second-order polynomial, which enables prediction of the effective index at a given wavelength. The model is constructed using a Taylor expansion around the central wavelength:

$$n_{\text{eff}}(\lambda) = n_1 + n_2 (\lambda - \lambda_0) + n_3 (\lambda - \lambda_0)^2 \quad (1)$$

Using the simulated data from Figure 4 and Figure 6, the Taylor expansion parameters can be extracted for TE and TM modes (Table II).

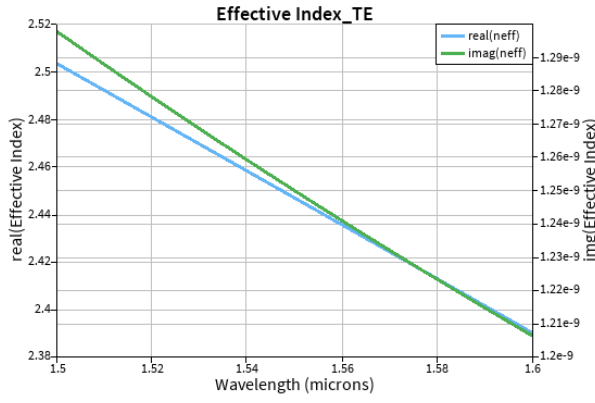


Fig. 4. Relationship of effective index and wavelength for TE mode.

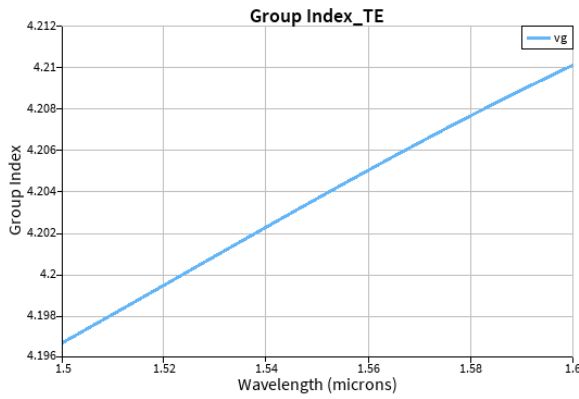


Fig. 5. Relationship of group index and wavelength for TE mode.

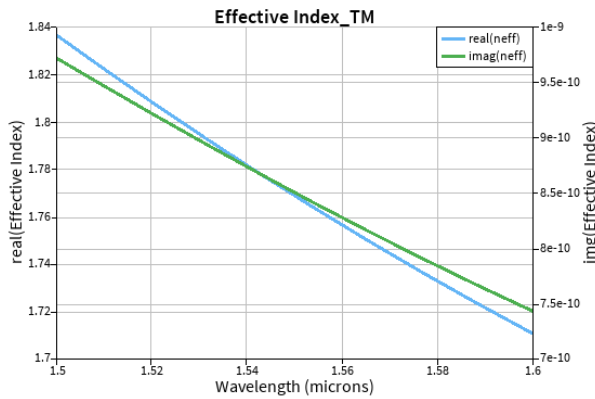


Fig. 6. Relationship of effective index and wavelength for TM mode.

TABLE I
TAYLOR EXPANSION PARAMETERS FOR COMPACT WAVEGUIDE MODEL

	n_1	n_2	n_3
TE	2.44682	-1.13340	-0.04398
TM	1.76882	-1.25870	1.91427

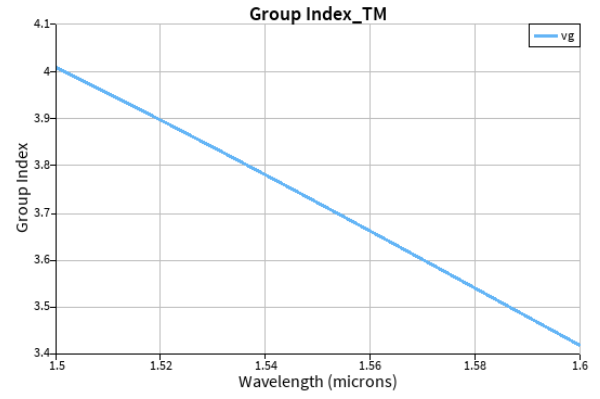


Fig. 7. Relationship of group index and wavelength for TM mode.

TABLE II
LIST OF DESIGN PARAMETERS FOR MACH-ZEHNDER INTERFEROMETER

ΔL [μm]	waveguide width [nm]	FSR_{TE} [nm]	FSR_{TM} [nm]
50	500	11.430	12.915
100	500	5.715	6.458
200	500	2.858	3.229
500	500	1.143	1.292

B. Mach-Zehnder Interferometer

The transfer function of the interferometer versus wavelength can be expressed as:

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta_1 L_1 - \beta_2 L_2)] \quad (2)$$

where I_i is the input light intensity, I_o is the output intensity, and L_1 and L_2 are lengths of the two waveguide arms, and β_1 and β_2 are the waveguide propagation constants, and are defined as:

$$\beta_{1,2} = \frac{2\pi n_{1,2}}{\lambda} \quad (3)$$

For an imbalanced Mach-Zehnder Interferometer with identical waveguide arms but an optical path length difference between these arms (ΔL), Equation 2 becomes:

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)] \quad (4)$$

The free spectral range (FSR) of this kind of imbalanced Mach-Zehnder Interferometer can be expressed as:

$$\text{FSR} = \frac{\lambda^2}{\Delta L n_g} [m] \quad (5)$$

Table II lists the design parameters for the imbalanced Mach-Zehnder Interferometer and expected FSR for TE and TM modes.

Figure 8 shows the transmission spectra of a Mach-Zehnder Interferometer with a path length difference of 100 μm . The blue curve shows the transmission spectrum without grating couplers, which has approximately zero insertion loss and flat wavelength response. By adding input and output grating couplers in the circuit, the green curve shows a more realistic

simulation where a wavelength dependent insertion loss is present in the transmission spectrum.

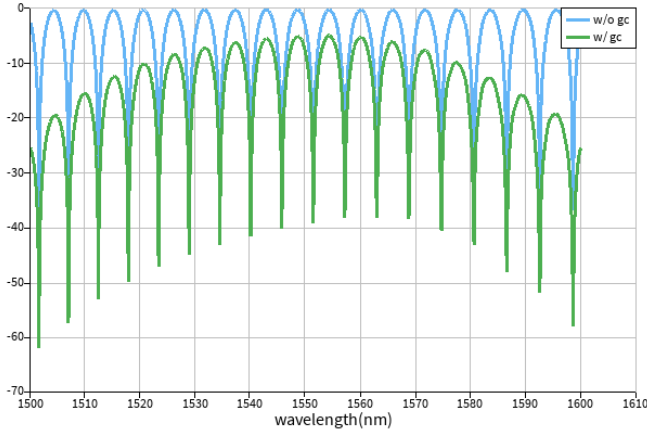


Fig. 8. Transmission spectra [dB] of a Mach-Zehnder Interferometer with $\Delta L = 100 \mu\text{m}$ with and without grating couplers.

III. LAYOUT

The final layout submitted for fabrication is shown in Figure 9, containing three MZIs for TE mode, two MZIs for TM mode, de-embedded structures for TE and TM modes, and a ring resonator. Note that some of the design parameters listed in Table II are not included in the layout due to limited area ($605 \mu\text{m} \times 410 \mu\text{m}$).

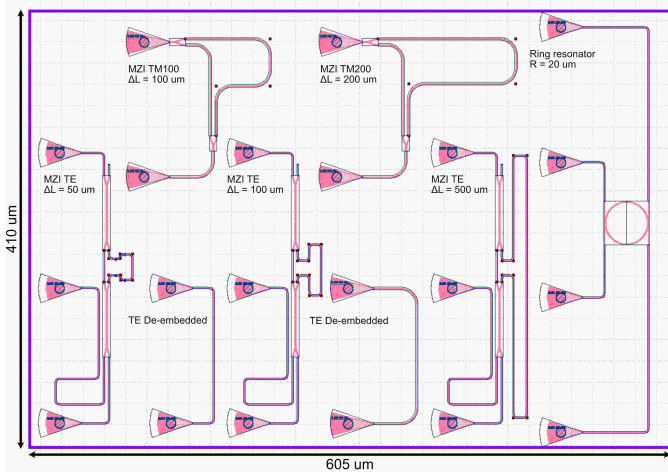


Fig. 9. Layout for fabrication.

IV. CORNER ANALYSIS

To account for manufacturing variations, corner analysis is performed at four corners with varying waveguide heights and widths for TE mode using Lumerical MODE. Effective index and group index are plotted for four corners and nominal design (Figure 10 and Figure 11). Using the simulated effective index values, five compact models are generated and the corresponding waveguide parameters are calculated for TE

TABLE III
COMPACT MODELS AND CALCULATED CONVENTIONAL PARAMETERS FOR THE WAVEGUIDES

W [nm]	L [nm]	n_1	n_2	n_3	n_{eff}	n_g	D [s/m^2]
500	220	2.45	-1.13	-0.04	2.45	4.20	4.54E-4
470	215.3	2.37	-1.21	-0.03	2.37	4.25	3.51E-4
470	223.1	2.41	-1.20	-0.05	2.41	4.26	5.48E-4
510	215.3	2.44	-1.12	-0.03	2.44	4.17	3.44E-4
510	223.1	2.47	-1.11	-0.05	2.47	4.19	4.79E-4

mode, at 1550 nm (Table III). The effective index ranges from 2.37 to 2.47 and the group index ranges from 4.17 to 4.26 at 1550 nm. Finally, these waveguide parameters are used in Lumerical INTERCONNECT to generate plots for the expected MZI transfer functions (Figure 12 and Figure 13).

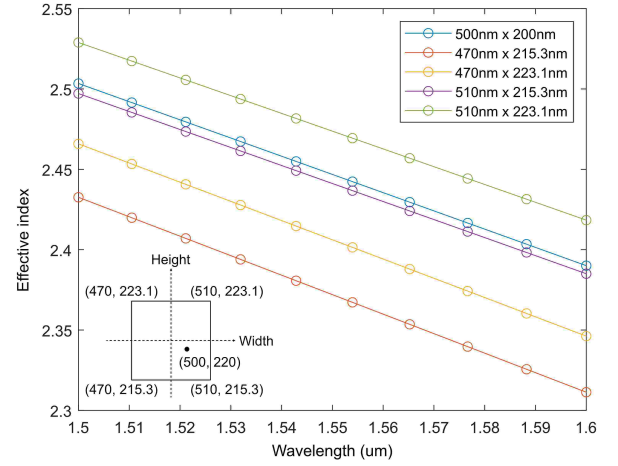


Fig. 10. Effective index of the TE mode for a waveguide with different width and height.

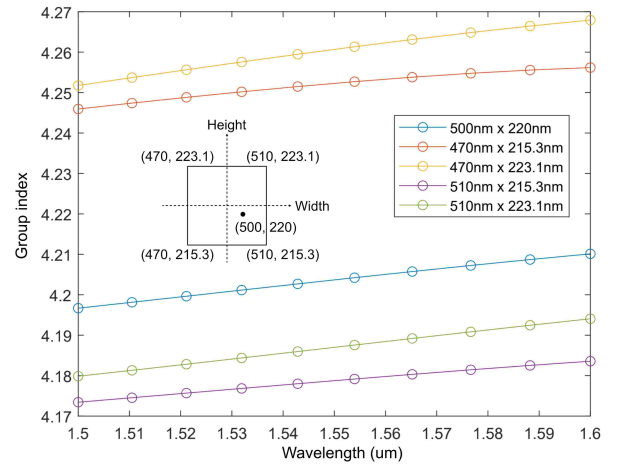


Fig. 11. Group index of the TE mode for a waveguide with different width and height.

V. EXPERIMENTAL DATA

All eight devices in the layout were measured. However, the discussion will focus on the MZIs for TE mode. Figure 14-

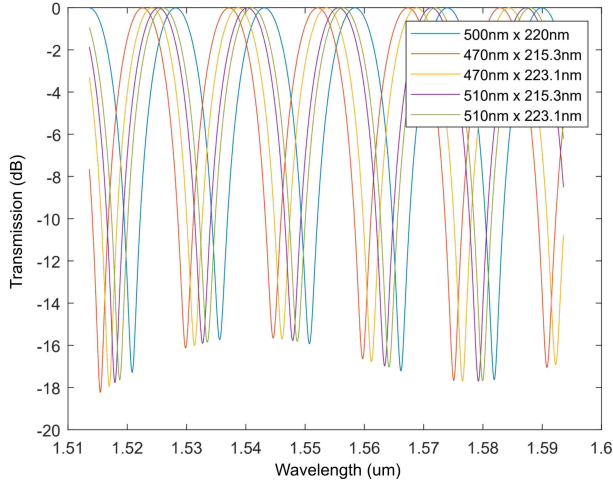


Fig. 12. Expected MZI transfer function with $\Delta L = 50 \mu\text{m}$.

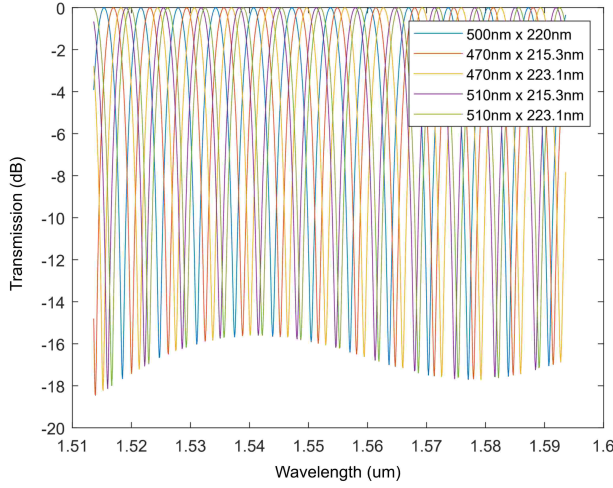


Fig. 13. Expected MZI transfer function with $\Delta L = 100 \mu\text{m}$.

16 show the transmission spectrum together with baseline fit for MZIs with $\Delta L = 50 \mu\text{m}$, $100 \mu\text{m}$, $500 \mu\text{m}$, respectively. As ΔL increases, the FSR becomes narrower, which agrees well with the simulation results (Table II). Measurement on the ring resonator is also included; however, this is beyond the scope of this report (Figure 17).

VI. DATA ANALYSIS

To extract the waveguide parameters from the measurement, the data is first flattened by subtracting the baseline and limited the wavelength range of interest (Figure 18). Then, it is fit to the MZI transfer function:

$$F = 10\log_{10}\left(\frac{1}{4}\left|1 + \exp\left[-i\frac{2\pi n_{eff}}{\lambda}\Delta L - \alpha\Delta L/2\right]\right|^2\right) + b \quad (6)$$

where α is the waveguide propagation loss and b is the excess insertion loss. Expression for effective index is a Taylor expansion around the central wavelength, as shown in Equation 1. Curve fitting is performed using MATLAB

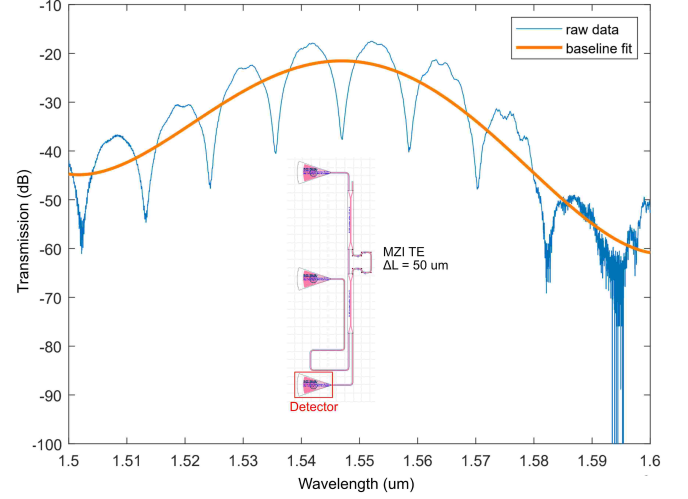


Fig. 14. Measurement on MZI for TE mode with $\Delta L = 50 \mu\text{m}$ and baseline fit.

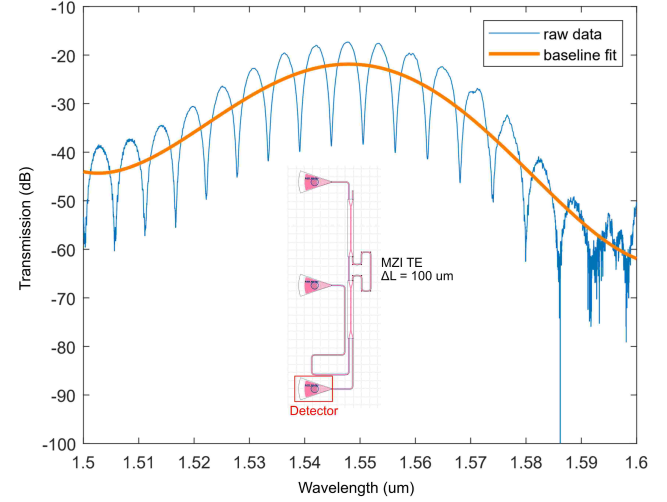


Fig. 15. Measurement on MZI for TE mode with $\Delta L = 100 \mu\text{m}$ and baseline fit.

lsqcurvefit function. By providing sufficiently close initial parameters, a successful fit can be achieved and the fitted parameters can be used to calculate the waveguide parameters using the given equations:

$$n_{eff} = n_1 \quad (7)$$

$$n_g = n_1 - n_2 \cdot \lambda_0 \quad (8)$$

$$D = -2 \cdot \lambda_0 \cdot n_3/c \quad (9)$$

Curve fitting is performed for MZIs for TE mode with $\Delta L = 50 \mu\text{m}$, $100 \mu\text{m}$, $500 \mu\text{m}$ (Figure 19-21). The extracted fitted parameters and calculated waveguide parameters are listed in Table IV. The estimated effective index and group index for these three MZIs fall within the range discussed in Section IV.

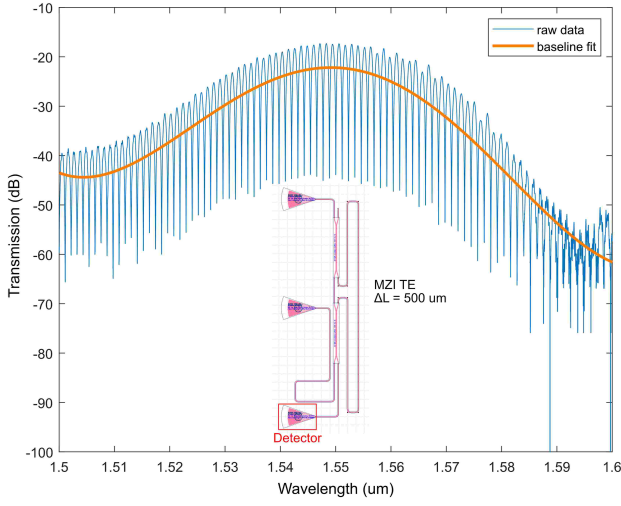


Fig. 16. Measurement on MZI for TE mode with $\Delta L = 500 \mu\text{m}$ and baseline fit.

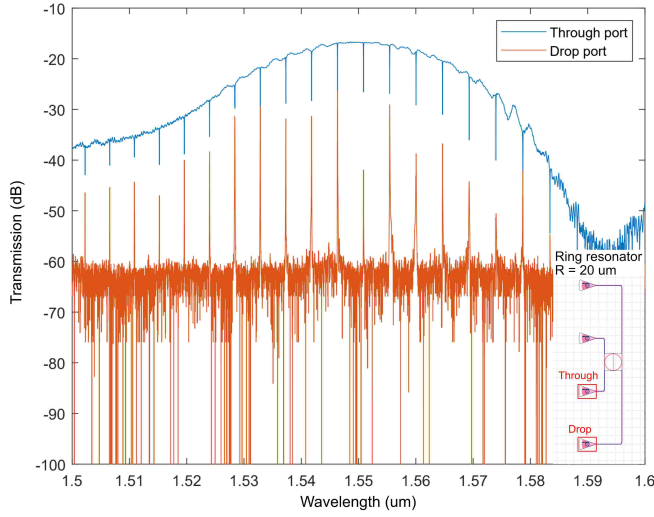


Fig. 17. Measurement on ring resonator with $R = 20 \mu\text{m}$.

TABLE IV
ESTIMATED PARAMETERS FROM CURVE FITTING

ΔL [μm]	n1	n2	n3	α	b	n_g	FSR [nm]
50	2.41	-1.14	0.19	0.0079	2.167	4.17	11.51
100	2.40	-1.15	-0.01	0.0037	1.547	4.18	5.746
500	2.45	-1.12	-0.06	0.0004	0.476	4.19	1.147

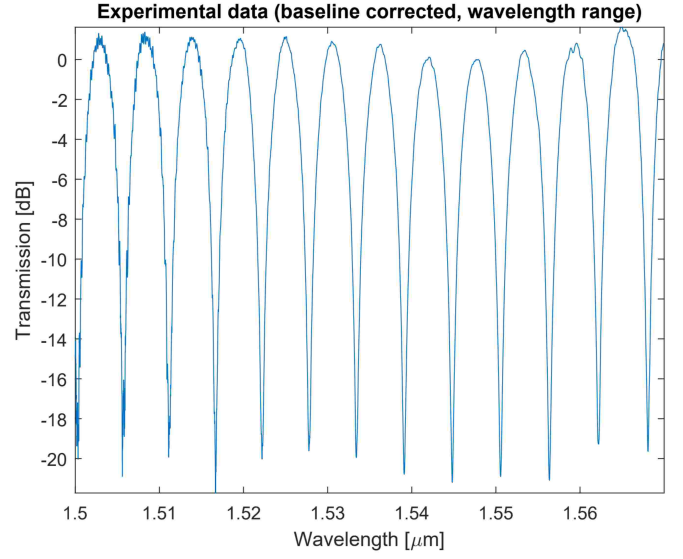


Fig. 18. Transmission spectrum of MZI for TE mode with $\Delta L = 100 \mu\text{m}$, baseline subtracted, and within wavelength range of interest.

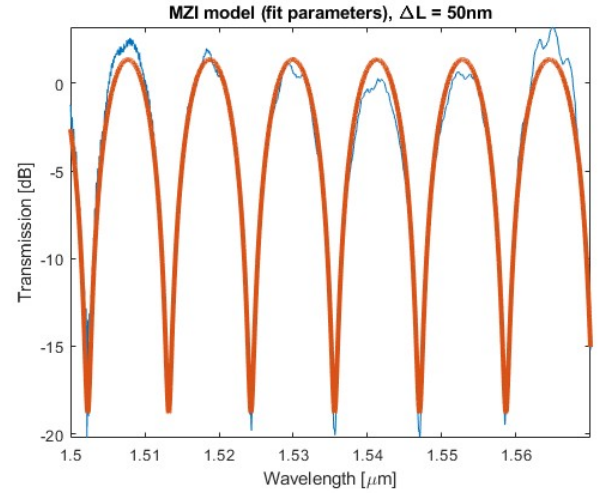


Fig. 19. Transmission spectrum and MZI model for TE mode with $\Delta L = 50 \mu\text{m}$.

VII. WAVEGUIDE TEMPERATURE DEPENDENCE

The MZIs for TE mode were measured at two different temperature points (25°C and 30°C), which can be used to estimate the thermo optic coefficient for the silicon waveguide. Curve fitting is performed at these two temperature points (Figure 22), and the corresponding waveguide parameters are extracted (Table V). Finally, the thermo optic coefficient can be calculated: $\frac{dn_{eff}}{dT} = 1.8 \times 10^{-4} \text{K}^{-1}$.

VIII. CONCLUSION

In this work, silicon photonic MZIs were designed, simulated, fabricated, and characterized. The measurement results show good agreement with the simulation. By fitting the measured data to the MZI transfer function, a few key waveguide

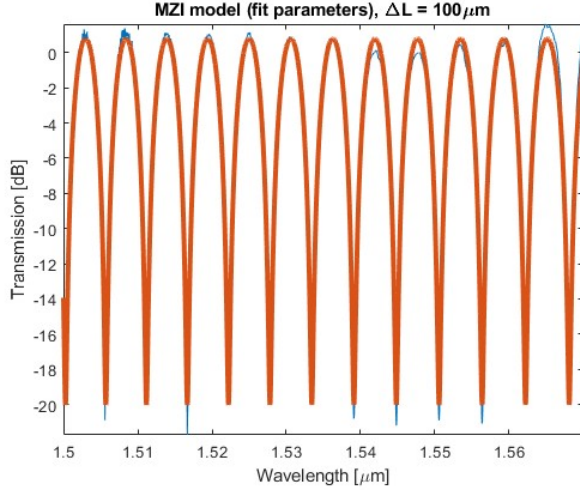


Fig. 20. Transmission spectrum and MZI model for TE mode with $\Delta L = 100 \mu\text{m}$.

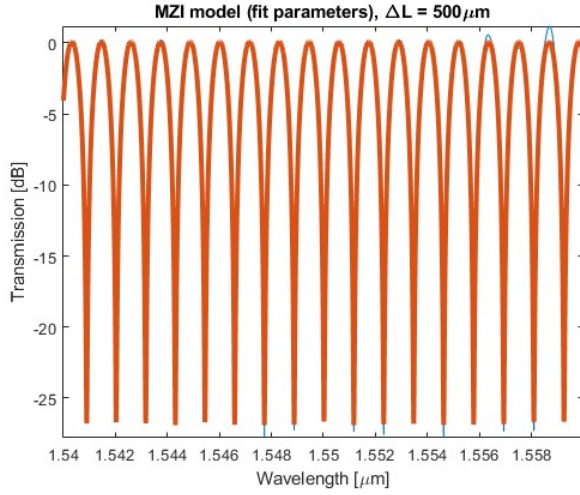


Fig. 21. Transmission spectrum and MZI model for TE mode with $\Delta L = 500 \mu\text{m}$. Note that the wavelength is limited to a narrower range to ease the difficulty for curve fitting.

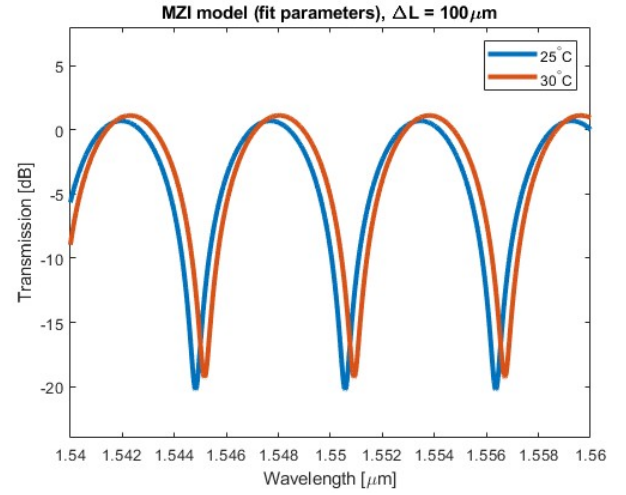


Fig. 22. MZI models for TE mode with $\Delta L = 100 \mu\text{m}$, measured at 25°C and 30°C .

parameters were successfully extracted, e.g. effective index, group index, propagation loss, dispersion, and thermo optic coefficient. Overall, this study demonstrates the effectiveness of MZI structures for characterizing silicon waveguides and validates the complete design-to-measurement workflow for silicon photonic integrated circuits.

TABLE V
ESTIMATED WAVEGUIDE PARAMETERS AT 25°C AND 30°C

Temperature [$^\circ\text{C}$]	n_{eff}	n_g	FSR [nm]
25	2.4020	4.1693	5.76234
30	2.4029	4.1697	5.76174